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PALEOMAGNETIC RESULTS FROM THE CENTRAL SIERRA NEVADA: CONSTRAINTS ON RECONSTRUCTIONS OF THE WESTERN UNITED STATES

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Abstract. Paleomagnetic results from three plutons of the central Sierra Nevada limit the motion of the Sierra Nevada relative to the North American craton to $5^{\circ} \pm 8^{\circ}$ of poleward motion and $7^{\circ} \pm 11^{\circ}$ of clockwise rotation since about 90 Ma. The indicated rotation and translation are not significant at the 95% confidence level. A new tectonic model is presented that is consistent with all available paleomagnetic data from the Sierra Nevada, Great Valley, Klamaths, Oregon Coast Range, Cascades, and western Nevada. The paleogeographic reconstructions of this model associate the displacement of major crustal blocks with Basin and Range extension starting at about 30 Ma. This model accounts for about 15° of clockwise rotation of the Klamaths and southern Cascades, about 2° of clockwise rotation of the Sierra Nevada and Great Valley, and 26° of clockwise rotation of the Oregon Coast Range since 30 Ma. The model is consistent with 40° of clockwise rotation of the Oregon Coast Range before 30 Ma, as proposed by Magill and Cox (1980). The amount of Basin and Range extension in this model at latitude 41°N is 225 km directed E-W and at latitude 35°N is 245 km directed WSW-ENE.

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INTRODUCTION

The two major tectonic processes which have deformed the western margin of North America during the Tertiary are extension in the Basin and Range province and right lateral shear associated with relative motion between the Pacific and North American plates. Many of the paleomagnetic poles determined from rock units in western North America indicate the northward motion and the clockwise rotation with respect to the North American craton expected as a consequence of dextral shear [Beck, 1976, 1980]. Basin and Range extension probably began in the Oligocene and has continued to the present (Thompson and Burke [1974], Wright [1976], Eaton et al. [1978], Lachenbruch and Sass [1978], Zoback and Thompson [1978], Eaton [1979], Gans [1982], Wernicke et al. [1982], and many others). Though the magnitude of extension within the Basin and Range province is a much-debated topic, the large spatial dimensions of the extending region coupled with its interior position with respect to the western margin of the North American plate lead to the conclusion that deformation of the western margin was a necessary consequence of extension [Hamilton and Myers, 1966]. Basin and Range extension was incorporated into a model of Magill and Cox [1980] and Magill et al. [1981] to explain the post-Eocene clockwise rotation of the Cascade Range and Oregon Coast Range inferred from paleomag-

netic data. In their model those two ranges together with the Klamath Mountains moved as a single quasi-rigid crustal block during the late Tertiary in conjunction with Basin and Range extension. The Sierra Nevada moved as a separate block. A hinge point at the junction of the Sierra Nevada and Klamath Mountains moved further toward the west than points to the north and south on both blocks, producing 27° clockwise rotation of the Klamaths-Oregon Coast Range-Cascade block and 15° counterclockwise rotation of the Sierra Nevada block. They argued that the 95% confidence limits determined in earlier paleomagnetic studies [Gromme and Merrill, 1965; Gromme et al., 1967; Mankinen, 1978; Hannah and Verosub, 1980] were great enough to permit as much as 15° of counterclockwise rotation of the Sierra Nevada. The present study of the Sierra Nevada was undertaken to test the validity of the proposed 15° counterclockwise rotation of the Sierra Nevada.

GEOLOGIC SETTING

The position of the Sierra Nevada, west of the expanding Basin and Range province and east of the major dextral shear zone, provides an opportunity to test the easterly extent of dextral shear and the geometric consequences of Basin and Range extension. The Sierra Nevada batholith is composed predominantly of granitoids 210 to 80 m.y. old [Evernden and Kistler, 1970] that were emplaced at depths of several kilometers [Kistler et al., 1971]. Comparison of zircon U-Pb ages with biotite and hornblende K-Ar ages generally shows good agreement for the younger granitoids [Stern et al., 1981], indicating that the K-Ar ages of the younger granitoids have not been reset significantly by regional reheating.

The study area in the Tower Peak quadrangle (Figure 1) was chosen because of the good exposures, the freshness of the rocks in this glaciated area, the availability of radiometric ages, and the availability of detailed geologic mapping [Wahrhaftig, 1984]. K-Ar ages of rocks in the Tower Peak area (Figure 1) show that the latest plutonic event in this area took place around 96-84 Ma and was widespread. Inclusions in the Grace Meadow pluton yielded a K-Ar biotite age of 85.0 ± 1.5 Ma and three K-Ar hornblende ages of 91.6 ± 1.2, 94.2 ± 2.4 and 96.2 ± 1.0 Ma (B. Turrin, personal communication, January 1983). K-Ar ages by Kistler [1973] for the Mount Gibson pluton in the adjacent Hetch Hetchy quadrangle, where the fine- and coarse-grained facies were not differentiated, are 85.3 Ma (biotite) and 83.6 Ma (hornblende), both corrected to new decay constants. A single biotite K-Ar age for the fine-grained Mount Gibson unit sampled in Tower Peak quadrangle is 93.5 ± 1.5 Ma (B. Turrin, personal communication, January 1983).

FIELD AND LABORATORY TECHNIQUES

Three hundred and sixty-four oriented samples were collected at 49 sites in the three Upper Cretaceous plutons shown in Figure 1. Nineteen sites were in diorite and gabbro inclusions in the Grace Meadow pluton (Figure 2a). Twenty-three sites were collected in the fine-grained and seven in the coarse-grained Mount Gibson pluton, a hornblende-biotite quartz diorite which consists of a western coarse grained unit that intrudes an eastern fine grained unit (Figures 1 and 2b).

During a pilot study, oriented hand samples were collected from three sites in the Grace Meadow inclusions and from three sites in the fine-grained Mount Gibson pluton. Cores drilled in the laboratory from these samples were oriented with an accuracy of ≃5°. At the remaining 43 sites, core samples were drilled in the field and oriented by using a sun compass or back azimuths with an estimated accuracy of 2°. From one to three specimens were prepared from each sample.

Pairs of specimens were selected from samples from 47 sites, one specimen for thermal and one for magnetic demagnetization experiments. For the latter, the specimen was demagnetized using progressive alternating fields with amplitudes ranging from 5 to 80 mT. The results plotted on vector component diagrams [Zijderveld, 1967] show linear demagnetization trends (Figure 3a), indicating the removal of a single component of magnetization. A second specimen from each of the 47 samples was stepwise thermally demagnetized by using a Schonstedt TSD-1 shielded furnace. Most samples exibited a narrow range of blocking temperatures with more than half of the NRM being lost in the range of ≃550°C to ≃600°C (Figure 3b). Some samples however, had wider ranges of blocking temperatures which is probably a grain-size effect. For these samples, some lost as much as 80% of the NRM at

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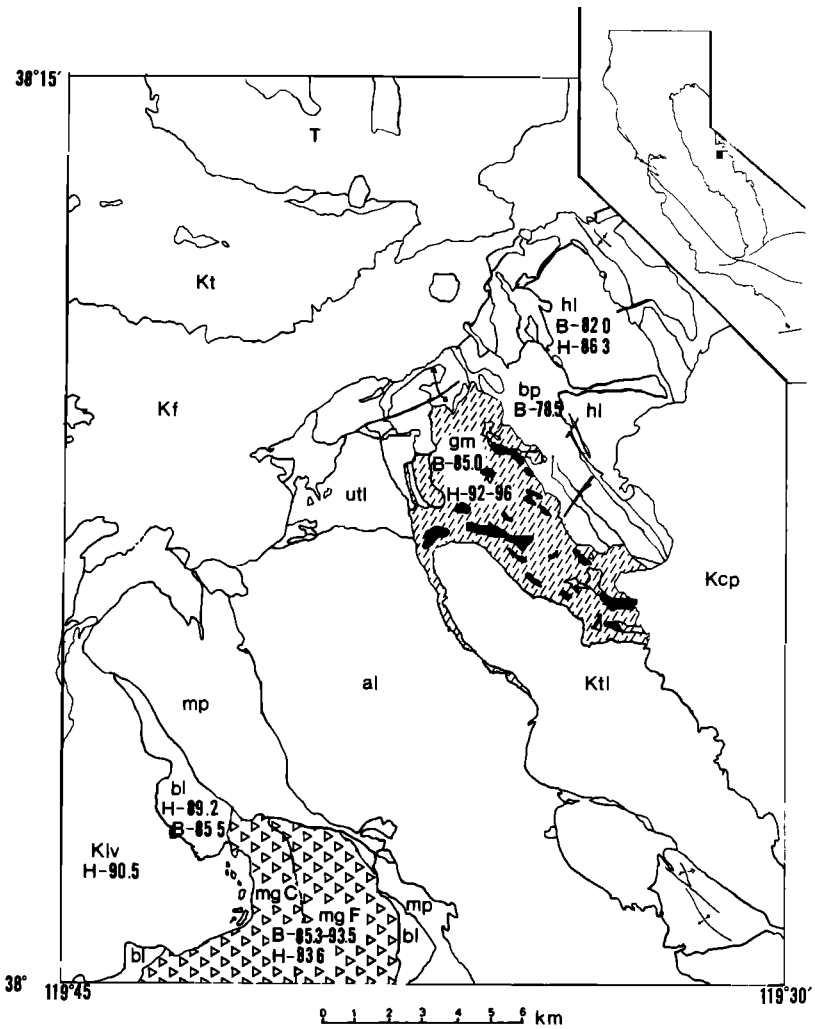


Fig. 1. The three plutons sampled in the Tower Peak quadrangle are the fine-grained Mount Gibson pluton (mgF), coarse-grained Mount Gibson (mgC) and Grace Meadow (gm). Mafic intrusions in the latter are shown on the map as dark areas. Also shown on the map are other Cretaceous plutons (formation symbols Kt, Kf, hl, bp, utl, Kcp, Ktl, al, mp, bl, Klv) and Tertiary rocks, mainly volcanics (symbol T). Areas not labeled are small plutons or formations of metamorphic rocks. Geologic map is after Wahrhaftig [1984]. K-Ar biotite (B) and hornblende (H) ages were determined by B. Turrin (personal communication, January 1983) and by Kistler [1973]. Ages from the latter were recalcuated using new decay constants.

temperatures below 550°C. The remaining magnetization was lost at temperatures between ≃550°C and ≃600°C. The characteristic directions determined by the two procedures were compared statistically and found to be the same at the 95% confidence level. Therefore magnetic cleaning was used for all but two sites, one in a Grace Meadow inclusion and one in the fine-grained Mount Gibson pluton, where more than one magnetic component was detected during thermal cleaning. All specimens from these two sites were thermally cleaned prior to statistical analysis.

PALEOMAGNETIC RESULTS

Approximately three quarters of the samples were found upon thermal and magnetic demagnetization to have stable, sin-

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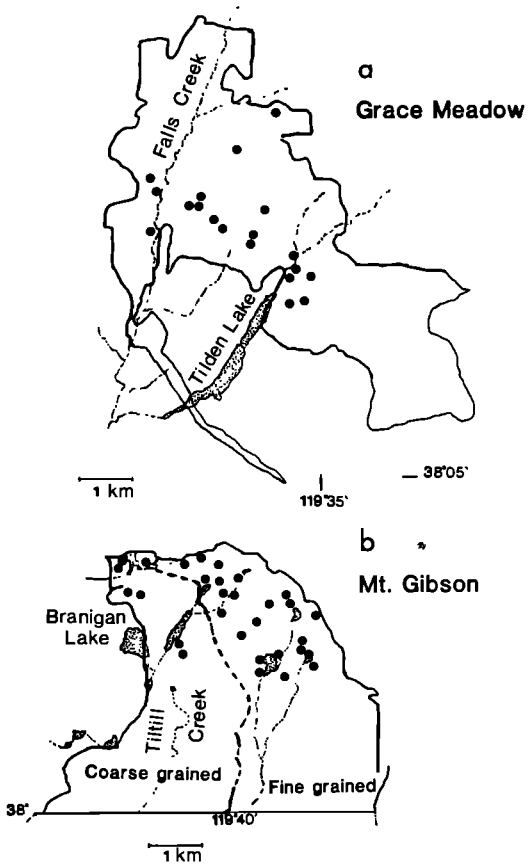


Fig. 2. Site locations (solid circles). In the Mount Gibson pluton, fine- and coarse-grained facies are separated by heavy dashed line. Stippled areas are lakes.

gle component magnetization, as evidenced by vector component plots that trended linearly to the origin. About one quarter showed a characteristic direction during stepwise alternating field (AF) demagnetization which, though remarkably straight, missed the origin by several percent of the original natural remanent magnetization (NRM) (Figure 3a). Thermal demagnetization of a second specimen from the same sample invariably showed that the small residual high-coercivity component had a low blocking temperature (Figure 3b), suggesting that it is a low-temperature overprint.

In stepwise isothermal remanence (IRM) experiments, specimens typically acquired saturation IRM in fields of 0.2-0.5 T, indicating that magnetite is probably the main ferromagnetic mineral present. NRM blocking temperatures, as mentioned above, indicate that magnetite is the main remanence carrier. The carrier of the low-temperature component was not identified.

Specimens from several sites were exceptions to this pattern of simple, stable magnetization. At several sites the NRM of one sample diverged markedly from that of other samples from that site both in direction of magnetization and in intensity, which was anomalously high by about an order of magnitude. Data from these specimens, which had probably been remagnetized by lightning, were rejected. In addition, five specimens from one of the Grace Meadow inclusion sites were unstable and showed, upon thermal demagnetization, a reversed low-temperature overprint over a normal component with high blocking temperatures due presumably to magnetite.

The polarities of all specimens from 47 of the 49 sites were normal. Samples from two of the sites in Grace Meadow inclusions, including the site with the specimens with unstable magnetic directions, showed antipolar reversed and normal components in the same sample: a low temperature reversed component with high coercivity overprinting a high temperature normal component with moderate coercivity (Figure 3c).

The observed ages in the range 96-84 Ma together with the stable normal magnetization seen in all specimens suggest that the plutons cooled near the end of the long Cretaceous normal period that lasted from 118 to 83 Ma (time scale of Harland et al. [1982]). The reversed polarity overprint is consistent with alteration in the late stages of cooling, possibly during the period of reversed polarity from 83 to 79 Ma.

ANALYSIS OF DATA

A characteristic direction was determined for each specimen after AF magnetic cleaning (47 sites) or thermal cleaning (two sites with low temperature overprints). These directions were combined using Fisher statistics to form site means (Tables 1, 2, and 3 and Figure 4), which were then combined to yield the mean paleomagnetic directions of the three rock units. Two means are given, one including data from all sites, and one excluding data from hand samples and from drilled sites with $\alpha_{95} > 9^{\circ}$. The means based on the winnowed data are used in subsequent discussions.

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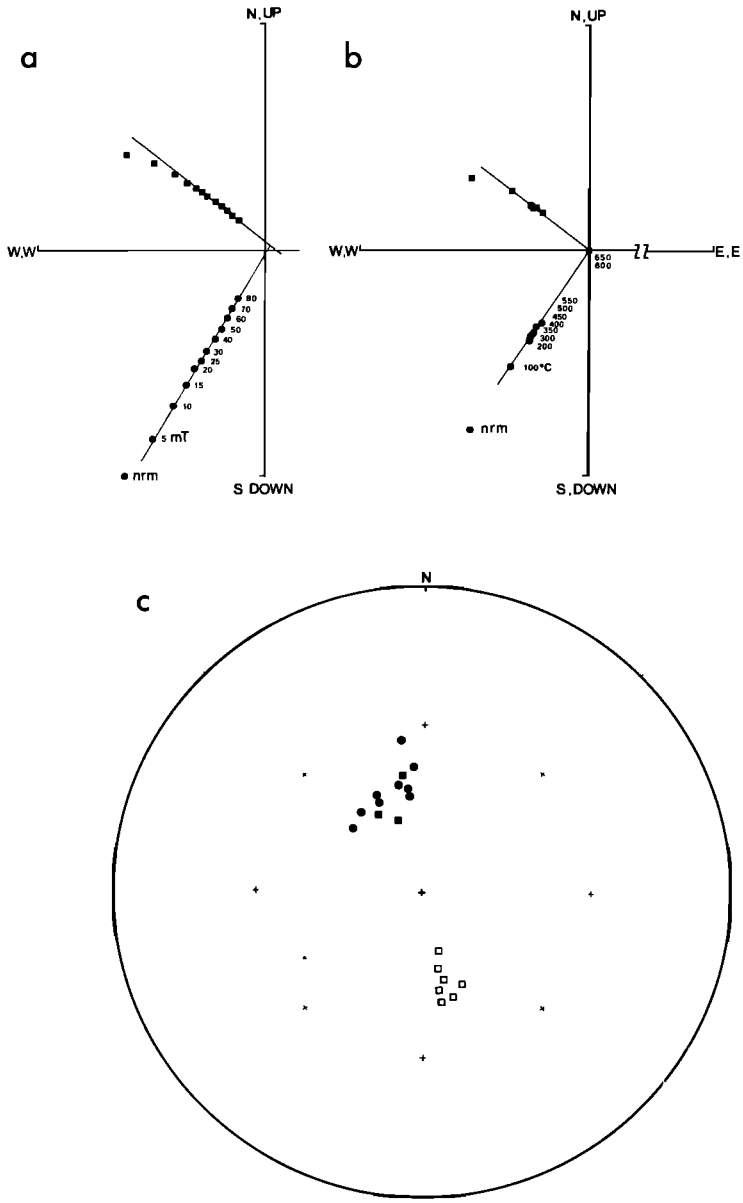


Fig. 3. Vector components after partially demagnetizing two specimens from one core of fine-grained Mount Gibson granodiorite are shown above. (a) AF demagnetization, length of axes 2.16 x 10^{-5} Am^2. (b) Thermal demagnetization, length of axes 1.96 x 10^{-5} Am^2. Squares: N and W components projected onto horizontal plane. Circles: down and west components projected onto E-W vertical plane. (c) Magnetization of samples from two sites. Circles: site H178 from fine-grained Mount Gibson pluton, where only normal magnetization is present. Squares: site H419 from a Grace Meadow inclusion, where both normal and reversed components are present. Open symbols: upper hemisphere. Solid symbol: lower hemisphere.

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TABLE 1. Site Mean Directions - Inclusions in Grace Meadow Pluton

Site	In Situ		Corrected		N	R	κ	α_{95}	Site Poles	
	Inc.	Dec.	Inc.	Dec.					N Lat.	E Long.
H187	65.3	7.0	65.4	4.8	8	7.671	21	12.3	79.9	259.1*
H195	60.3	11.9	60.5	10.2	7	6.953	129	5.3	81.5	303.6
H203	-63.9	155.3	-63.5	153.5	3	2.994	324	6.9	69.1	178.4
H211	67.2	329.4	66.7	327.4	8	7.892	65	6.9	64.2	186.6
H221	69.3	337.2	68.9	334.8	7	6.958	139	5.1	67.4	197.7
H229	75.6	335.4	75.2	332.0	9	8.960	198	3.7	60.4	214.0
H238	67.8	331.1	67.3	329.0	12	11.928	151	3.5	64.9	189.1
H253	68.5	330.9	68.0	328.7	7	6.979	296	3.5	64.5	191.1
H271	62.7	337.9	62.3	336.1	7	6.918	73	7.1	71.2	175.1
H400	69.0	346.5	68.7	344.0	7	6.855	42	9.5	72.1	206.9*
H409	61.7	337.5	61.3	335.8	10	9.776	40	7.7	71.1	171.3
H419	-65.1	165.8	-64.8	163.7	7	6.975	240	3.9	75.2	191.5
H419(N)	65.1	343.3	64.8	341.3	3	2.979	96	12.6	73.7	188.6***
H427	61.9	354.4	61.8	352.5	6	5.932	74	7.8	82.5	193.7
H263	66.4	331.0	65.9	329.0	7	6.939	98	6.1	65.4	184.7
H280	70.5	358.7	70.5	355.9	8	7.935	108	5.4	73.1	232.2
H435	60.7	356.2	60.6	354.4	8	7.947	132	4.8	84.4	191.5
H006	65.9	329.0	65.4	327.1	4	3.875	24	19.2	64.3	182.6**
H011	58.3	315.3	57.6	314.2	4	3.978	134	8.0	54.4	165.2**
H015	70.3	314.3	69.6	312.4	4	3.973	110	8.8	54.2	191.4**
All sites	66.6	340.7	66.3	338.5	19	18.833	108	3.3	71.0	192.3
All sites but *,**,***	66.4	343.1	66.1	341.0	14	13.904	135	3.4	72.5	193.9

Site poles are corrected for tilt.

In situ: field directions uncorrected for tilt. Corrected: field directions after correction for 1° of tilt to the west. Inc.: Inclination. Dec.: Declination. N: number of samples or sites averaged. R: vector sum. κ: precision parameter. α_{95}: radius of 95% confidence in degrees. Lat.: latitude. Long.: longitude.

*Sites with $\alpha_{95} > 9^{\circ}$.

**Sites from pilot study hand samples.

***The normal components found in 3 samples by vector subtraction after partial thermal demagnetization; not included in mean.

Within each of the three plutons the between-site dispersion of site poles is remarkably small. The value of κ expected at this latitude for virtual geomagnetic poles is 22, based on global paleomagnetic measurements of geomagnetic secular variation during the past 5 m.y. [Harrison, 1980]. The observed between-site values within the three plutons are 58, 57, and 173. Part of this dispersion is due to true between-site dispersion and part to errors in the site means. Using site α_{95} values to estimate the errors in the site means gives estimates of 70, 71, and 286 for the κ values of the true between-site dispersion. One possible explanation for the low dispersion is that all sites in each pluton cooled through their blocking temperatures in a time interval shorter than that required to average out secular variation, roughly 10^4 to 10^5 years. This seems unlikely given the dimensions of the plutons (5-10 km) and the vertical and horizontal distribution of the sampling locations within the plutons (Figure 2). Because of the slow cooling times of plutons this size and because a range of blocking temperatures was observed, the more likely explanation of the small intraunit angular dispersion is intrasite and intraspecimen averaging of geomagnetic secular variation.

The mean field directions of the two Mount Gibson intrusive units are not significantly different from each other. However, at the 95% confidence level, both

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TABLE 2. Site Mean Directions - Fine-Grained Mount Gibson Pluton

Site	In Situ		Corrected		N	R	κ	α_{95}	Site Poles	
	Inc.	Dec.	Inc.	Dec.					N Lat.	E Long.
H100	64.3	333.8	63.8	332.0	10	9.931	131	4.2	68.0	179.0
H111	62.7	346.3	62.4	344.4	8	7.935	107	5.4	76.9	181.0
H119	63.5	341.5	63.2	339.6	7	6.970	202	4.3	73.4	180.4
H127	61.0	345.7	60.7	344.0	8	7.953	148	4.6	77.2	172.0
H136	66.2	307.2	65.4	305.9	9	8.890	73	6.1	50.0	182.0
H145	51.5	335.0	51.1	333.9	8	7.901	71	6.6	67.8	141.9
H154	60.1	319.2	59.4	317.9	8	7.893	66	6.9	57.6	167.6
H162	59.2	304.9	58.4	304.0	7	6.919	74	7.1	46.9	170.0
H170	60.5	338.5	60.1	336.9	7	6.982	334	3.3	72.0	167.1
H178	61.4	341.8	61.1	340.1	9	8.907	86	5.6	74.3	171.8
H444	65.0	347.2	65.0	347.2	7	6.924	79	6.8	77.0	198.0
H453	60.0	344.9	59.7	343.2	7	6.944	108	5.8	76.8	166.5
H461	64.5	331.7	64.0	329.9	9	8.950	161	4.1	66.5	179.0
H539	56.5	345.0	56.2	343.6	8	7.910	77	6.3	76.9	149.7
H547	47.7	333.9	47.3	332.9	8	7.877	57	7.4	65.5	135.4
H557	54.1	320.5	53.5	319.5	8	7.860	50	7.9	57.3	155.9
H567	59.4	332.7	58.9	331.2	7	6.892	55	8.2	67.6	163.5
H301	59.1	340.2	58.7	338.6	8	7.857	49	8.0	73.3	161.8
H309	66.0	329.5	65.5	327.6	7	6.943	106	5.9	64.6	183.0
H317	56.0	351.0	55.8	349.5	8	7.641	20	12.9	81.5	142.0*
H019	57.1	302.6	56.3	301.8	4	3.900	30	17.0	44.6	167.7**
H023	47.8	326.6	47.2	325.7	3	2.946	37	20.6	59.9	141.9**
H064	57.6	331.3	57.1	329.9	3	2.984	127	11.0	66.3	158.9**
All sites	59.8	332.6	59.4	331.2	23	22.749	88	3.3	67.8	165.9
All sites but *,**	60.7	333.8	60.3	332.3	19	18.831	106	3.3	68.5	168.4

Notes as in Table 1.

TABLE 3. Site Mean Directions - Coarse-Grained Mount Gibson Pluton

Site	In Situ		Corrected		N	R	κ	α_{95}	Site Poles	
	Inc.	Dec.	Inc.	Dec.					N Lat.	E Long.
H470	60.1	340.1	59.7	338.5	8	7.963	190	4.0	73.2	165.7
H479	61.3	346.7	61.1	344.9	9	8.933	119	4.8	77.8	174.8
H489	57.7	350.4	57.5	348.8	6	5.950	101	6.7	81.2	154.3
H497	63.2	348.9	63.0	347.0	6	5.991	552	2.9	78.3	187.8
H514	52.2	335.2	51.8	334.0	8	7.924	93	5.8	68.1	143.4
H522	53.9	340.6	53.6	339.3	8	7.904	73	6.5	72.9	143.5
H530	61.7	340.0	61.3	338.3	8	7.991	800	2.0	72.9	172.0
All sites	58.7	342.8	58.4	341.2	7	6.976	253	3.8	75.4	161.1

Notes as in Table 1.

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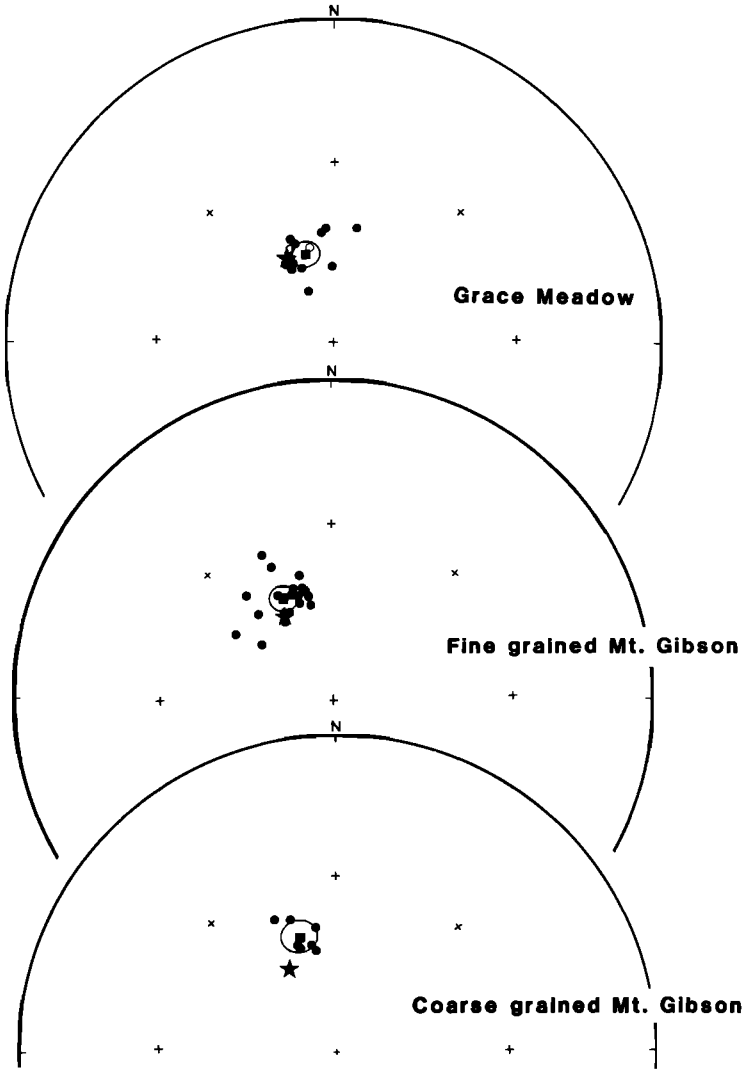


Fig. 4. Site mean directions of magnetization after magnetic cleaning. Solid circles: site mean directions for sites with normal polarity. Open circles: site mean directions for sites with reversed polarity, transposed 180°. Squares: pluton mean directions (means of site means) with corresponding circle of 95% confidence. Stars: expected directions calculated from Irving and Irving [1982]. In situ magnetic directions are plotted on the lower hemisphere of an equal area projection.

are significantly different from the result from the Grace Meadow pluton. Three possible explanations for the observed difference are (1) inadequately averaged secular variation, (2) shifting of the geomagnetic pole between the times of magnetization of the plutons, or (3) tectonic tilt or rotation of one pluton relative to the other. The first is unlikely for reasons discussed earlier. The second seems unlikely because the pole path between the mean Grace Meadow pole and the Mount Gibson poles does not trend in the same direction as do the polar wander curves of Harrison and Lindh [1982] and Irving and Irving [1982]. Moreover, the latter's mean rate of polar wander from 100 to 80 Ma is 0.14 degrees/m.y., which would produce only 2° of polar displacement in rocks with ages varying over the observed range in ages, whereas the observed divergence in poles is 9°.

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Tectonic movement of the plutons along major faults is unlikely because no such faults between the Mount Gibson and Grace Meadow plutons can be seen in this well exposed area [Wahrhaftig, 1984]. Although joints are common in the study area and some display slickensides, little or no displacement is apparent [C. Wahrhaftig, personal communication, 1983]. On the other hand, pervasive micro-faulting as described by Lockwood and Moore [1979] throughout the Sierra Nevada could, under some geometrical conditions, cause differential rotations. It seems likely that the cause of the slight discordance in magnetic directions between plutons is structural, but the exact mechanism of deformation is not known.

No basis for making a local structural tilt correction was available at any of the sites. As in most studies of plutons, the validity of the final result rests on the degree of consistency between widely separated sites and on the assumption that whatever tilting has occurred is random from site to site and not systematic throughout the region sampled. So far as is known, the latter assumption is justified for the Sierra Nevada except for a regional tilt of the entire Sierra Nevada block of about 1° to the west [Christensen, 1966; Slemmons et al., 1979]. This tilt correction is made in Tables 1, 2, and 3 and Figure 5.

The small but statistically significant differences that exist among the mean directions of magnetization of the three plutons could be recongnized only as a result of the extensive sampling. Had the sampling been less extensive and therefore these differences not recognized, the statistical analysis would be a mean of the 40 site means as N = 40 independent observations, obtaining a 95% confidence circle of 3.1° (Table 4). However, having recognized small but significant differences in the magnetic directions of the plutons, the correct procedure is to calculate the final paleomagnetic pole from the mean of the three pluton means, giving a more conservative but also more realistic 95% confidence circle of 9.4°. Our pole at 72.6°N and 174.7°E is not significantly different at the 95% confidence level from the results of the early paleomagnetic study of Grommé and Merrill [1965] of Late Cretaceous plutons in the Sierra Nevada with a pole at 69°N and 195°E and an A_{95} of 9.6° based mainly on NRM data with no magnetic cleaning.

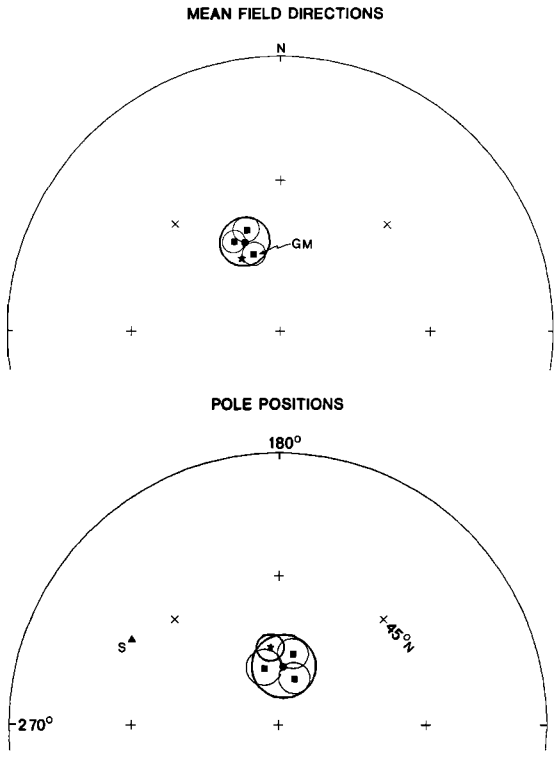


Fig. 5. Mean paleomagnetic directions and poles for the three plutons. Solid squares: means and 95% confidence circles (light ovals) for the plutons. GM refers to Grace Meadows. Solid circle: mean of the pluton means and 95% circle of confidence (heavy oval). Star: result expected at 90 Ma on basis of paleomagnetic data from North America craton [Irving and Irving, 1982] and 95% confidence circle (heavy oval). Triangle: sampling site location (38.05°N, 240.35°E). Data are tectonically corrected for 1° westward tilt of the Sierra Nevada. Projections are lower hemisphere equal area.

Under the assumption that the paleomagnetism of the sampling area is chartacteristic of the Sierra Nevada crustal block, the amount of clockwise rotation of the Sierra Nevada relative to the craton was calculated. The expected declination, calculated on the basis of paleomagnetic data from the craton, was subtracted from the observed paleomagnetic declination. For the expected direction three alternative cratonal poles (Table 5) were used to give the following three values for the rotation, r, and its 95% confidence interval, Δr: $6.5^{\circ} \pm 11.2^{\circ}$, $6.8^{\circ} \pm 10.8^{\circ}$, and $2.6^{\circ} \pm 14.0^{\circ}$. Simi-

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TABLE 4. Mean Pole Positions

Rock Unit	N Lat.	E Long.	N	R	κ	A_{95}
Grace Meadow inclusions	72.5	193.9	14	13.777	58	5.3
Fine-grained Mt. Gibson pluton	68.5	168.4	19	18.686	57	4.5
Coarse-grained Mt. Gibson pluton	75.4	161.1	7	6.965	173	4.6
(All sites combined)	(71.6)	(175.6)	(40)	(39.284)	(55)	(3.1)
Three plutons combined	72.6	174.7	3	2.988	173	9.4

Notes as in Table 1.

larly, the amount of post-Cretaceous north-northwesterly displacement of the Sierra Nevada crustal block in a direction toward the location of the Late Cretaceous pole was calculated. The paleocolatitude expected in the Late Cretaceous on the basis of paleomagnetic data from the craton was subtracted from our new paleomagnetic determination of the paleocolatitude of the Sierra Nevada. The three alternative cratonal poles (Table 5) give the following values for the poleward displacement, d, and its 95% confidence interval, Δd: $5.1^\circ \pm 8.0^\circ$, $4.9^\circ \pm 7.7^\circ$, and $6.5^\circ \pm 9.6^\circ$. (Equations for calculating the rotation, the poleward displacement and their 95% confidence limits [Cox, 1980] are given in Table 5.) The amounts of rotation and poleward displacement found by using the three alternative cratonal poles are not significantly different. Since the 90 Ma North American pole of Harrison and Lindh [1982] includes earlier paleomagnetic data from the Sierra Nevada, the subsequent analysis will be done using the values found from the 90 Ma North American pole of Irving and Irving [1982].

DISCUSSION

Constraints on Paleogeographic Reconstructions Imposed by the Paleomagnetic Data From the Sierra Nevada

Possible Late Cretaceous positions of the sampling locality consistent with the mean paleomagnetic inclination lie along the small circle L (Figure 6), which is concentric around the 90 Ma expected pole. The 95% confidence limits for the 90 Ma position of the sampling site are shown in Figure 6 by the two small circles L_1 and L_2. Because the present day position of the sampling site is included within these limits, the paleomagnetic results are consistent with the hypothesis that the Sierra Nevada has undergone no poleward displacement. However, the present results do not preclude as much as 13° of poleward displacement. The 5° of displacement corresponding to the mean Sierra

TABLE 5. Poleward Displacement d and Clockwise Rotation r Calculated for Different Expected Poles

References for the Expected Pole	d*	Δd*	r*	Δr*
Irving and Irving [1982] 90 m.y.	5.1	8.0	6.5	11.2
Harrison and Lindh [1982] 90 m.y.	6.5	9.6	2.6	14.0
Mean Late Cretaceous Pole**	4.9	7.7	6.8	10.8

*$d = p_o - p_e$ (p_o is observed paleocolatitude and p_e is distance from sampling site to the expected cratonal pole). $\Delta d = (\Delta p_o^2 + \Delta p_e^2)^{1/2}$ is 95% confidence limit for d, where $\Delta p_o = 0.78\ A_{95}^o$ and $\Delta p_e = 0.78\ A_{95}^e$. ($A_{95}^o$ is observed A_{95} and A_{95}^e is expected A_{95}). $r = D_o - D_e$ (D_o is observed declination and D_e is expected declination). $\Delta r = (\Delta D_o^2 + \Delta D_e^2)^{1/2}$ is 95% confidence limit for r, where $\Delta D_o = 0.78 \sin^{-1}(\sin A_{95}^o / \sin p_o)$ and $\Delta D_e = 0.78 \sin^{-1}(\sin A_{95}^e / \sin p_e)$. Multiplication by 0.78 is after Demarest (1983).

** Mean pole at 66.9°N, 185.5°E with $A_{95} = 3.2^\circ$, $\kappa = 1841$, N = 4 is based on the following poles selected from the literature: Hanna [1967, 1973], Larochell and Black [1963], Larochelle et al. [1963], Scharon and Hsu [1969] and Shive and Frerichs [1974].

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Nevada paleomagnetic inclination would not, in fact, be anomalous viewed in the context of the widespread occurrence in western North America of northward displacements (Beck [1976] and Table 5) which took place in the early Tertiary tectonic environment of subduction and shear.

The amount of clockwise rotation found in the present study (Table 5), $7^{\circ} \pm 11^{\circ}$, is not consistent with the 15° of counterclockwise rotation required by the model of Magill and Cox [1980]. Although the observed 95% confidence limits are consistent with no net clockwise rotation or even with a 4° counterclockwise rotation, the most likely rotation is the 7° clockwise rotation calculated from the mean declination.

The observation that the Sierra Nevada is presently rotated clockwise relative to the craton does not rule out the possibility of one or more intervals of post-Cretaceous counterclockwise rotation. If the Sierra Nevada moved to its present position by rotation about a fixed rotation pole, then all rotation was clearly clockwise. However, if the style of motion was more complex, intervals of counterclockwise rotation may well have alternated with intervals of clockwise rotation. Because the Sierra Nevada block is located along the western boundary of the Basin and Range province, the extensional history of which was quite complex, it appears unlikely that the westerward motion of the Sierra Nevada occurred by simple rotation about a fixed rotation pole.

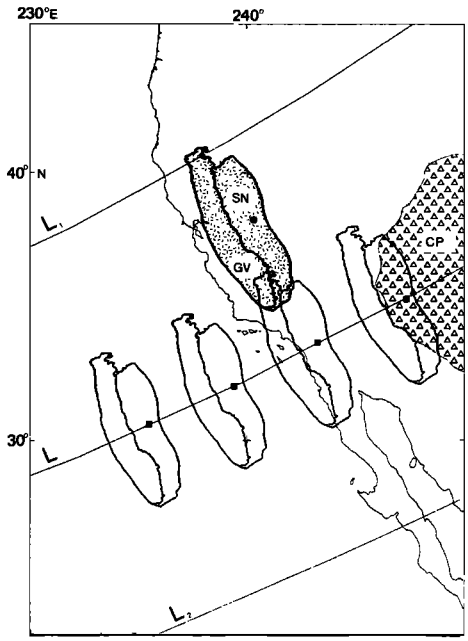


Fig. 6. Sierra Nevada (SN) and Great Valley (GV) is shown by stippled pattern and position of sampling site by solid circle. Possible Late Cretaceous positions of the Sierra Nevada lie along the small circle L, which is concentric about the Late Cretaceous cratonal paleomagnetic pole with a radius determined by the mean paleomagnetic inclination. Small circles L_1 and L_2 correspond to the limits of 95% confidence for the reconstructed position of the Sierra Nevada and because the sampling site lies between them, the paleomagnetic data are consistent with no poleward displacement of the Sierra Nevada. The Colorado Plateau (CP) is shown with figured patterns in its present position.

Additional Geological and Paleomagnetic Observations that Constrain Paleogeographic Reconstructions

Published paleomagnetic results from the Oregon Coast Range, the Cascades, the Klamaths, the Great Valley, the Sierra Nevada, and western Nevada are summarized in Table 6. Several models that attempt to explain the paleomagnetic results from the Oregon Coast Range and the Cascades have been proposed. Magill et al. [1981] suggested a two stage model for the rotation of the Oregon Coast Range. Phase I of the rotation, active between 50 and 42 Ma, is a result of accretion of the Oregon Coast Range. Phase II, active between 20 Ma and present, is associated with Basin and Range extension. In an alternative model Beck and Plumley [1980] propose steady rotation at a rate of about 1.5°/m.y. of the Oregon Coast Range from early Eocene until early Miocene (50-20 Ma). Recent paleomagnetic results from the Klamath block [Mankinen and Irwin, 1982], the California Cascades [Craig and Beck, 1981], southern Washington [Sheriff and Bentley, 1980; Wells et al., 1983], northwestern Nevada [Russell et al., 1982] and the Sierra Nevada [this study] suggest that some revisions of the reconstruction models are appropriate.

Large quasi-rigid crustal blocks. In order to relate the paleomagnetically measured rotations to the geologic history of this area, some initial assumptions must be made concerning the style of deformation and the dimensions of the

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TABLE 6 - Poleward Displacement d and Clockwise Rotation r for the Oregon Coast Range, the Cascades, the Klamaths, the Great Valley, the Sierra Nevada and Western Nevada Compared to Cratonal Poles From Irving and Irving [1982]

Rock Type or Locality	r ***	r	Δr	d	Δd	Age	Reference
Oregon Coast Range							
Tyee-Flournoy	74	68	± 13	4	± 9	Eocene	1
Siletz River volc.	74	72	± 10	8	± 7	Eocene	1
Tillamook	26-74	45	± 9	5	± 6	Eocene	2
Yachats Basalts	26-74	54	± 23	0	± 16	Eocene	1
Eocene Intrusions	26-74	53	± 17	4	± 12	Eo-Oligocene	3
Oligocene Intrusions	26	45	± 9	8	± 7	Oligocene	3
Miocene Intrusions	9-26	35	± 21	-6	± 13	Oligo-Miocene	3
Cascades							
Oregon western Cascades N*	26	26	± 20	-11	± 11	Oligocene	4
Oregon western Cascades S*	15	27	± 25	-4	± 16	Oligocene	4
California western Cascades	15	15	± 8	2	± 6	Oligocene	5
Klamath							
Hornbrook**	15	8	± -	0	± -	Upper Cretaceous	6
Great Valley							
Great Valley	2	6	± 8	3	± 5	Upper Cretaceous	7
Sierra Nevada							
Reeve Formation	2	5	± 25	-3	± 20	Upper Jurassic	8
Bucks Batholith	2	-11	± 20	-15	± 14	Lower Cretaceous	9
Cretaceous Granites	2	4	± 13	-3	± 8	Upper Cretaceous	10
Yosemite plutons	2	7	± 11	5	± 8	Upper Cretaceous	11
Northwestern Nevada							
Black Rock Desert region	-	1	± 11	0	± 7	Upper Cretaceous	12

Notes as in Table 5.
* Northern and southern sites averaged separately.
** Samples with primary magnetization compared to 90 m.y. pole.
***r' is the theoretical paleomagnetic declination anomaly, as explained in the text. For rock units with ages that are between two frames of the model, a range is given.
References numbered as follows: 1-Simpson and Cox [1977]; 2-Magill et al. [1981]; 3-Beck and Plumley [1980]; 4-Magill and Cox [1980]; 5-Craig and Beck [1981]; 6-Mankinen and Irwin [1982]; 7-Mankinen [1978]; 8-Hannah and Verosub [1979]; 9-Grommé et al. [1967]; 10-Grommé and Merrill [1965]; 11-this study; 12-Russell et al. [1982].

blocks that have rotated. If the dominate mechanism of deformation was regional penetrative shear, then pervasive faulting on a small scale might be expected to produce rotated blocks with dimensions ranging from a few centimeters to perhaps a few kilometers. Alternatively, if quasi-rigid blocks with dimensions of several hundred kilometers rotated with little internal deformation, then most regional deformation would be confined to areas between the blocks. Though our paleogeographic reconstructions will be based on the assumption that the blocks behave quasi-rigidly, we do not mean to imply that the blocks are unbroken by faults or the absence of all deformation internal to the blocks. The assertion is that deformation and faulting internal to the blocks is substantially less than that occurring between the quasi-rigid blocks. Magill and Cox [1980] proposed that the Klamath, Cascade and Sierra Nevada mountain ranges comprise such quasi-rigid blocks, pointing out that these relatively undeformed blocks are adjacent to regions of greater structural mobility characterized by extensive Pliocene and Pleistocene normal faulting and high heat flow (inset, Figure 7). Geologic and geophysical evidence

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which suggests the Oregon Coast Range is also a quasi-rigid block includes the continuity of outcrop of middle Eocene Tyee and Flournoy formations, the absence of strong deformation of these formations, the consistency throughout the region of current directions and paleomagnetically determined rotations, and the low spatial frequency of the gravity field [Magill and Cox, 1980; Magill et al., 1981]. Southern Washington appears to constitute a distinctly different tectonic regime in which different tectonic blocks have undergone different amounts of rotation [Wells and Coe, 1980; Sheriff and Bentley, 1980; Bates et al., 1981; Magill et al., 1981; Wells et al., 1983].

Depositional contacts of the Oligocene western Cascade volcanics onto both the Klamath Mountains and the Oregon Coast Range led Magill and Cox to consider the Cascades, the Oregon Coast Range, and the Klamaths as one block. However, recent paleomagnetic data from Oligocene volcanics of the southern Cascades, including western Cascade volcanics in California deposited on the flank of the Klamath block (Table 6), now suggest that this southern area rotated less than the Oregon Coast Range and northern Cascades. Thus our current preferred reconstruction includes three quasi-rigid blocks: the Oregon Coast Range and northern Cascades block, the Klamath Mountains and southern Cascades block, and the Sierra Nevada and Great Valley block. The present position of these blocks is shown in Figure 7.

Role of extension. Because these three quasi-rigid blocks comprise the moving western boundaries of regions of Basin and Range extension, geologic information about the location, direction, and timing of the extension plays a crucial role in our paleogeographic models. Extension has been taking place east of the Sierra Nevada since the Oligocene. The early stages of extension now exposed in metamorphic core complexes occur in a belt from Mexico to Canada [Coney, 1979; Armstrong, 1982]. South of the Snake River plain this extension took place in the Oligocene-Miocene, and was directed NW-SE in the northern Basin and Range province (for location see Figure 7), and NE-SW in the southern Basin and Range province (Davis [1980], Davis et al. [1980], Wernicke [1982], Gans and Miller [1983], and more). Dike swarms and fault slip vectors indicate WSW-ENE extension in the period 20-10 Ma throughout the Basin and

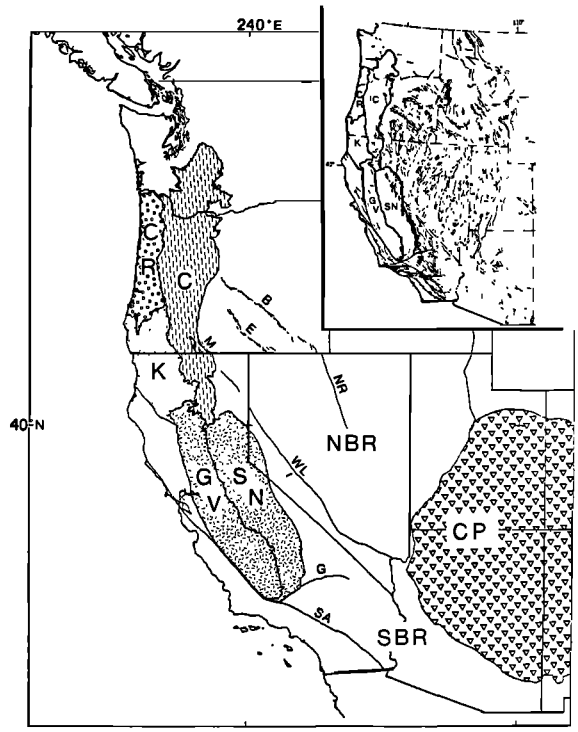


Fig. 7. Our model divides the western United States into two tectonic regimes, one with intense extensional deformation and the other comprised of quasi-rigid crustal blocks. The regions of extension are the Northern and Southern Basin and Range province (NBR and SBR). The quasi-rigid crustal blocks are the Sierra Nevada-Great Valley (SN & GV) block, the Oregon Coast Range-Northern Cascades (CR & C) block, and the Klamath-Southern Cascades (K & C) block. The approximate boundary between the northern and southern Cascades is shown in Figure 9. Faults are the Garlock (G), Walker Lane (WL), Nevada Rift (NR), San Andreas (SA), Brothers fault zone (B), Eugene-Denio fault zone (E), and Mclaughlin fault zone (M). The projection is mercator. In the inset, the present day position of the tectonic blocks is outlined with heavy lines. Light lines show steeply dipping faults active in the past 10-15 m.y. for which some Quaternary movement is suspected [Eaton, 1980].

Range [Zoback et al., 1981]. In the northern Basin and Range, WNW-ESE to E-W extension by block faulting took place from 10 Ma to present, while block faulting in southern Basin and Range took place 13-10 Ma, and the extension was oriented

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TABLE 7. Finite Rotation Poles Used to Reconstruct Crustal Blocks From Present Positions to Position at Age T

Crustal Block	Age T of Reconstruction	Finite Rotation Pole		
		N Latitude	E Longitude	CCW Rotation*
Sierra Nevada (SN)	10	33.0	239.0	-7.0
and Great Valley (GV)	30	65.87	225.82	4.47
	50	65.87	225.82	4.47
Oregon Coast Range (CR)	10	46.0	237.5	9.0
and northern Cascades (C)	30	46.0	237.5	27.0
	50	44.0	237.62	72.98
Klamath (K)	10	46.0	237.5	9.0
and southern Cascades (C)	30	47.82	237.51	17.02
	50	47.82	237.51	17.02

* Counterclockwise rotation in reconstruction.

WSW-ENE (Eaton et al. [1978], Zoback and Thompson [1978], Eaton [1979], Davis [1980], Zoback and Zoback [1980], Zoback et al. [1981], Eaton [1982], and others). Estimates for the amount of Basin and Range extension range from 20% to 30% [Stewart, 1978, 1980] to 100-300% in specific areas (Hamilton and Myers [1966], Wernicke et al. [1982], Gans and Miller [1983], and others].

Paleogeographic Reconstructions

Palinspastic reconstructions based on finite rotations of the three quasi-rigid crustal blocks described earlier were used to model the geologic information about Basin and Range extension and paleomagnetic information about rotations (Table 6). In discussing these models, rotations of two types will be distinguished. The first is rotation of a crustal block relative to North America about a finite rotation pole. This pole specifies the operation required to move the crustal block from its present position to its former position at the time of the reconstruction in coordinates that remain fixed relative to the craton.

The second type of rotation is the change in azimuth, r', of a hypothetical north arrow scribed upon a crustal block prior to its displacement as the block moves from its former position to its present position. Finite rotations of the first type are used to calculate rotations of the second type, which are sometimes described as the rotations of blocks about vertical axes through the sampling sites. (This usage is rather imprecise because rotation of a block about any pole on the globe other than the paleomagnetic pole will produce a nonzero value of r'.) The second type of rotation, r', is a theoretical paleomagnetic declination anomaly, that is to be compared with the paleomagnetic declination anomaly $r = D_o - D_e$, where D_o is the paleomagnetic declination observed in rocks of a given age and D_e is the declination expected at the sampling site on the basis of paleomagnetic results from rocks of the same age on the craton. The declination anomaly is a direct measure of the amount that the block has rotated.

The poles used to produce the palinspastic reconstructions are given in Table 7. They were found by a process of iteration in which many different reconstructions were made and a preferred one selected on the basis of consistency with paleomagnetic data and regional geology and minimization of overlaps and gaps between blocks except where these are supported by geological observations. The former positions of the crustal blocks found by using our preferred set of rotations are shown in Figure 8.

In fitting our reconstruction model to the paleomagnetic data, we tried to find rotation poles that would produce values of r' that would exactly match observed values of r, while at the same time satisfying relevant geological constraints. Where this was not possible, we tried to find reconstuctions that would satisfy the geological constraints and also produce values of r' within the 95% confidence

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Fig. 8. Paleogeographic reconstructions based on poles listed in Table 7. The amount of westward motion is shown by arrows. The calculated amount of clockwise rotation undergone by a given block from the position at the time shown to the present is listed as r' in Table 6. The pre-30 Ma rotation of the Oregon Coast Range is after Magill and Cox (1980).

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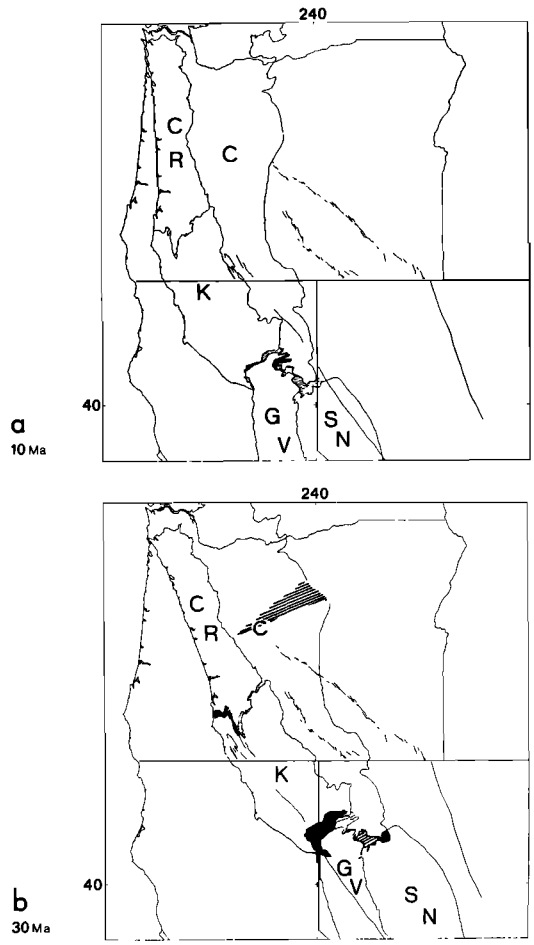


Fig. 9. Gaps that open between the present irregular boundaries between crustal blocks when the latter are rotated backward in time as quasi-rigid bodies are shown by diagonal shading. Regions where the blocks boundaries overlap in the reconstructions are shown in black. Beginning with the 30 Ma reconstruction and moving forward in time, the diagonal shading shows regions of expected compression and the solid pattern shows regions of expected extension.

interval of r. A comparison of observed values of r and values of r' calculated for our displacement model (Table 6) shows that the two are generally consistent.

Reconstruction at 10 Ma. The objective of this reconstruction is to provide a first order model for post 10 Ma extension in the northern Basin and Range province, as determined from geological and geophysical data measurements of the direction and magnitude of extension. Two crustal blocks were rotated to create this reconstruction, a Sierra Nevada-Great Valley block and a Cascades-Klamath-Oregon Coast Range superblock (Figure 8a, Table 7). These rotations restore 95 km of post 10 Ma WNW extension of the Basin and Range province at about latitude 41°N, tapering southward to no extension at 33°N, the latitude of the pole of the rotation. Because our reconstruction keeps the southern end of the Sierra Nevada at about its present distance from the Colorado Plateau, it does not account for the 43 km of post middle Pliocene left lateral displacement along the Garlock fault reported by Carter [1982]. Small overlaps and gaps appear in our 10 Ma reconstruction near the pivot points between the Sierra Nevada-Great Valley block and the Klamaths-Cascades-Coast Range block (Figure 9a). The overlaps, gaps, and the Garlock disagreement are a measure of the divergence of these blocks from the rigidity assumed in our model and should find geologic expression in the deformation of adjacent parts of the crustal blocks. Indications of recent deformation include the presence of young faulting in the northern Great Valley [Helley et al., 1981] and the presence of a zone of enhanced seismicity immediately north of latitude 40°N in the northern Sierra Nevada-Great Valley block [Zoback and Zoback, 1980].

Reconstruction at 30 Ma. During the interval 30-10 Ma the Sierra Nevada-Great Valley block, the Oregon Coast Range-Northern Cascades block, and the Klamath-Southern Cascades block all were displaced about different poles (Table 7). In the case of the Sierra Nevada-Great Valley block, we were unable to find a displacement with a rotation, r', which matched the observed declination anomaly, r = 7°, and which, at the same time, satisfied the geological constraints on the displacements of the northern and southern ends of the block. A 7° clockwise rotation of the Sierra Nevada block would require either a large displacement between the Sierra Nevada and Klamath blocks, which is unlikely because of the onlap of Late Cretaceous Great Valley sediments onto both blocks, or it would require placing the southern end of the Sierra Nevada adjacent to the Colorado Plateau. The latter would imply that the 100+ km width of basement presently exposed between the southern Sierra Nevada and the Colorado Plateau originated in a strip 10 km or so wide,

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which would require an unreasonably large amount of stretching. Our reconstruction moves the southern end of the Sierra Nevada to within 100 km of the Colorado Plateau and produces an apparent clockwise rotation of the Sierra Nevada block of r' = 2°, which lies well within the 95% confidence region of the paleomagnetically determined rotation, r = $7^{\circ} \pm 11^{\circ}$.

As noted earlier, Oligocene volcanic rocks of the western Cascades appear from paleomagnetic data to comprise two separate blocks, one in southern Oregon and northern California that has rotated 15° and one in northern Oregon that has rotated 26°. Volcanic rocks in southern Washington of similar age have undergone rotations comparable with those observed in northern Oregon [Beck and Burr, 1979; Bates et al., 1981], but as noted earlier the dimensions of rotated crustal blocks appear to be smaller in Washington than in Oregon [Magill et al., 1981, 1982]. In our reconstruction we have rotated the Oregon Coast Range and the northern Cascades as one block (r' = 26°) and the Klamaths and the southern Cascades as a second block (r' = 15°), in agreement with the paleomagnetically determined rotations. The net amount of Basin and Range extension compensated for by this model at latitude 41°N is 225 km directed E-W and at latitude 35°N is 245 km directed WSW-ENE.

Going backward in time from 10 to 30 Ma, our reconstruction produces a gap between the northern and southern western Cascade blocks because of the differential rotation between them (Figure 9b). Going forward in time, this gap becomes a zone of overlap corresponding to a region of compression that developed between 30 and 10 Ma. The location of the boundary between the northern and southern blocks is not well determined on the basis of available paleomagnetic data. Folds in Miocene rocks north of latitude 44°N, west of the axis of the Cascades and east of the Willamette River [Wells and Peck, 1961], suggest a possible locus for deformation between the two blocks and field evidence for deformation of the western Cascade volcanics in northern Oregon has been reported by J. G. Smith (personal communication, 1982). The boundary between the Oregon Coast Range block and the Klamath block proper is probably near the depositional contact of Eocene sediments upon Klamath basement, which is the boundary shown in Figure 7. The small gaps and overlaps produced in our reconstruction between the Oregon Coast Range and Klamath blocks (Figure 9) may be accommodated along some of the faults mapped near this contact.

Geologic observations preclude major Tertiary east-west motion between the Klamath and Sierra Nevada blocks because of the onlap of Upper Cretaceous Great Valley sediments onto both blocks. On the other hand, the presence of faults as young as Pleistocene in the northern Great Valley [Helley et al., 1981] indicates that some displacement has probably taken place between the two blocks. We have modeled these geologic observations by permitting modest (several tens of kilometers) but not large displacements between the Klamath and Sierra Nevada blocks. In our preferred model, gaps appear between the southern Cascades and Great Valley blocks and overlaps appear between the Great Valley and the Klamath blocks. These pose a serious problem for the model because they are not easily reconciled with local geology. A possible explanation is that the implied deformation is distributed over adjacent parts of the blocks, which would require relaxation of the assumption that blocks are perfectly rigid.

In the reconstruction from 30 to 50 Ma, the Sierra Nevada, Great Valley and Klamath blocks were kept in their 30 Ma positions and the Oregon Coast Range was rotated counterclockwise as suggested by Magill et al. [1981]. In our model the net post early Eocene rotation, r', of the Oregon Coast Range relative to North America is 74° clockwise, in accord with the paleomagnetically determined rotations (Table 6).

In these reconstructions, the amount of poleward displacement, d', of the Sierra Nevada during the past 30 m.y. is equal to 0.6°, less than the observed value of d = $5^{\circ} \pm 8^{\circ}$ but well within the error limits. If, as is likely, the amount of poleward displacement since the Cretaceous is greater than that of our reconstruction, this displacement may well have been part of a regional dextral shear that occurred prior to 30 Ma. If this displacement took place between 30 and 50 Ma, the reconstruction for 50 Ma (Figure 8c) should be revised. In this case the Oregon Coast Range, the Klamath Mountains the Great Valley and the Sierra Nevada should be displaced southwards relative to North America.

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This reconstruction is different from the one suggested by Magill and Cox [1980] in three respects:

1. In this model the Oregon Coast Range and the northern Cascades rotate as one block and the Klamaths and southern Cascades rotate separately by a smaller amount, whereas in the reconstruction of Magill and Cox [1980] the Klamath Mountains, Cascades, and Oregon Coast Range all rotate as one block.

2. In this model the Sierra Nevada and Great Valley rotate counterclockwise back to their position at 30 Ma, whereas in the reconstruction by Magill and Cox they rotate clockwise.

3. In this model Basin and Range extension begins at 30 Ma, whereas in the model by Magill and Cox [1980] it begins at 20 Ma. The amounts of extension at latitudes 41°N and 35°E are 225 km and 245 km, respectively, according to the present model and 340 km and 210 km, respectively, according to the model of Magill and Cox.

Our model based on three crustal blocks rotating without internal deformation about a simple set of rotation poles appears to be in agreement with most of the available paleomagnetic data and is grossly consistent with the regional geology. On the other hand, the overlaps and gaps produced in the reconstruction clearly indicate that at least along their margins the blocks have undergone some deformation. Moreover, the paleomagnetic study of Kanter and McWilliams [1982] and the structural study of Segall and Pollard [1983] demonstrate that parts of the Sierra Nevada block have undergone deformation. Perhaps the strongest conclusion that should be drawn from the consistency of our model with paleomagnetic and geological data is not that the three blocks have moved as perfectly rigid bodies, but rather, that much of the larger scale regional deformation has taken place in the region between the three blocks.

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